

Question 1: $P(A)=0.4$ $P(B)=0.5$ $P(A \cup B)=0.7$

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Section 2

$$1) P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.5 - P(A \cap B) = 0.7$$
$$= 0.9 - P(A \cap B) = 0.7$$

$$\boxed{P(A \cap B) = 0.2}$$

2) If A and B are mutually exclusive $\Rightarrow (A \cap B) = 0$

$P(A \cap B) = 0.2 \neq 0 \Rightarrow$ A and B are not mutually exclusive.

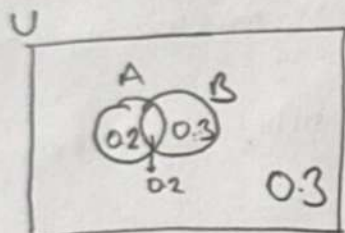
3) If A and B are independent $\Rightarrow P(A \cap B) = P(A) \times P(B)$

$$P(A)=0.4 \quad P(B)=0.5 \quad P(A \cap B)=0.2$$

$$0.4 \times 0.5 = 0.2$$

Since $P(A \cap B) = P(A) \times P(B)$, events A and B are independent.

4)



$$P(A' \cap B) = P((U-A) \cap B) = P(B-A) = 0.3$$

$$P(A' \cup B') = P((U-A) \cup (U-B)) = P(U - (A \cap B)) = 0.8$$

Question 2:

Law of Probability is $P(A) = \sum_{i=1}^n P(A|B_i) \cdot P(B_i)$ so,

$P(D) = 0.04 \rightarrow$ probability of that an item is defective

$P(\text{Test}/D) = 0.92 \rightarrow$ probability the test correctly identifies a defective item

$P(\text{Test}/nD) = 0.03 \rightarrow$ probability the test incorrectly marks non-defective item as

$P(nD) = 1 - 0.04 = 0.96 \rightarrow$ probability of non-defective item

defective

$$P(\text{test}) = P(\text{test}/D) \cdot P(D) + P(\text{test}/nD) \cdot P(nD)$$

$$= (0.92) \times (0.04) + (0.03) \times (0.96)$$

$$= 0.0368 + 0.0288$$

$$\boxed{P(\text{test}) = 0.0656 = \%6.56}$$

Question 2.2:

$$\text{Bayes' Theorem} = P(B_i | A) = \frac{P(A | B_i) \cdot P(B_i)}{P(A)} \quad \text{so;}$$

$$P(D | \text{test}) = \frac{P(\text{test} | D) \cdot P(D)}{P(\text{test})}$$

$$P(D | \text{test}) = \frac{0.92 \cdot 0.04}{0.0656} = 0.56097 \approx 0.561$$

$$P(D | \text{test}) = 0.561 = 56.1\%$$

Question 2.3:

- The test quite good at finding defective products → 92% probability correct detection.
- The probability that some of the sound products are mistakenly marked as defective (false rate 2.88)
- As a result, when the test is positive, the probability that the product is actually defective is 56.1%

So there is a nearly 44% rate that the test is incorrect, which indicates that the test is unreliable.

⇒ As a result;

It is good at detecting test defects.

But test result is not reliable

In that case we need the second test for tested items.

Question 3.1:

Total number of possible selections:

$$\text{Total} = \binom{20}{3} = \frac{20 \cdot 19 \cdot 18}{3 \cdot 2 \cdot 1} = 1140$$

number of wanted situation 8 (exactly 2 defective)

$$\text{Favorable} = \binom{15}{1} \times \binom{5}{2} = 15 \cdot \frac{5 \cdot 4}{2} = 150$$

$$P(\text{exactly 2 defective}) = \frac{\text{Favorable}}{\text{Total}} = \frac{150}{1140} = 0.1316 = 13.16\%$$

Question 3.2:

$$P(\text{all 3 defective | at least 1 defective}) = \frac{P(\text{all 3 defective})}{P(\text{at least 1 defective})}$$

$$P(A|L) = \frac{P(A \cap L)}{P(L)} = \frac{P(A)}{P(L)} \Rightarrow \text{Because } L \subset A \text{ so } P(A \cap L) = P(L)$$

$$P(A) = \frac{\binom{5}{3}}{\binom{20}{3}} = \frac{10}{1140} = \frac{1}{114}$$

$$P(L) = 1 - P(\text{all good}) = 1 - \frac{455}{1140}$$

$$P(\text{all good}) = \frac{\binom{15}{3}}{\binom{20}{3}} = \frac{15 \cdot 14 \cdot 13}{3 \cdot 2 \cdot 1} = \frac{455}{1140}$$

$$P(L) = 1 - \frac{455}{1140} = \frac{685}{1140}$$

$$\frac{P(A)}{P(L)} = \frac{1/114}{685/1140} = \frac{10}{685} \approx 0.0146$$

Question 3.3:

$$a) P(A) = \frac{\binom{5}{1}}{\binom{20}{1}} = \frac{1}{4} = 0.25$$

$$P(B|A) = \frac{\binom{4}{1}}{\binom{19}{1}} = \frac{4}{19} \approx 0.2105 \text{ (If first part is defective)}$$

$$P(B|A^c) = \frac{5}{19} \approx 0.2632 \text{ (If first part is not defective)}$$

we can use Law of Total Probability Theorem

$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$$

$$P(B) = \left(\frac{4}{19} \cdot \frac{5}{20}\right) + \left(\frac{5}{19} \cdot \frac{15}{20}\right) = \left(\frac{20}{380}\right) + \left(\frac{75}{380}\right) = \frac{95}{380} = 0.25$$

$$P(A) = 0.25$$

$$P(B) = 0.25$$

b) $P(A \cap B)$?

$$P(A \cap B) = P(A) \cdot P(B|A) = \frac{5}{20} \cdot \frac{4}{19} = \frac{1}{19} \approx 0.0526$$

$$\boxed{P(A \cap B) = 0.0526}$$

↪ If A and B are independent $\Rightarrow P(A \cap B) = P(A) \cdot P(B)$

$$P(A) \cdot P(B) = 0.25 \cdot 0.25 = 0.0625$$

$P(A \cap B) \approx 0.0526 \neq 0.0625$ so A and B are not independent.